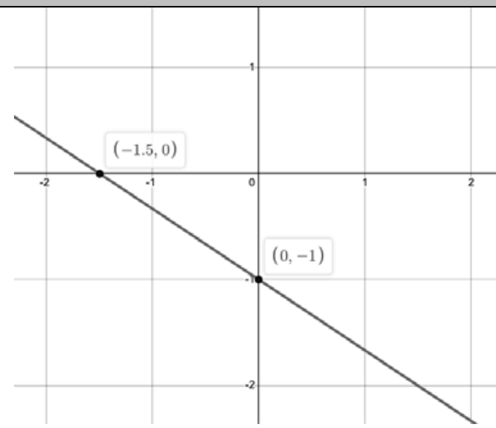
Algebra 1 Name____________________
Unit 4 Date____________________
Lesson 8 Homework Period__________________

For questions 1 through 4, use the table showing Lines A and B.

<table>
<tr><th>Line A</th><th>Line B</th></tr>
<tr><td>$B(x) = -\frac{2}{3}x + 3$</td><td>
</td></tr>
</table>

1. Which linear function has the larger slope?

2. Which line has the larger x-intercept?

3. Name all the key feature(s) that are the same for both linear functions.

4. Create a linear function that has the same y-intercept as Line A.

5. Complete the statements below to describe the end behavior of Line A.

   a. As $x \to \infty$, $y \to$
      - o $-\infty$
      - o ∞

   b. As $x \to -\infty$, $y \to$
      - o $-\infty$
      - o ∞

For questions 6 through 10, use the table showing Lines A and B.

<table>
<tr><th>Line A</th><th>Line B</th></tr>
<tr><td>A line passes through the points $(1, 9)$ and $(4, 15)$.</td><td><table><tr><th>x</th><th>y</th></tr><tr><td>-3</td><td>4</td></tr><tr><td>0</td><td>5</td></tr><tr><td>3</td><td>6</td></tr><tr><td>6</td><td>7</td></tr></table></td></tr>
</table>

6. Which linear function has the smaller slope?

7. Which linear function has the larger y-intercept?

8. Name the key feature(s) that are different for the two functions.

9. Create a linear function that has the same slope as Line B.

10. Consider the end behavior for the two functions.

    a. Complete the statement.

    Lines A and B have
    - o the same end behavior.
    - o different end behaviors.

    b. Write the end behavior of the functions.